

One-Dimensional Mouse-Trap Cascade Model

A step-by-step mean-field construction

1 Define the lattice

Consider a line of N sites. Each site independently contains one trap with probability ρ and is empty with probability $1 - \rho$. The expected initial number of traps is therefore

$$N_{\text{traps}} = N\rho. \quad (1)$$

The model may use open or periodic boundaries, but the choice must be stated.

2 Define the activation mechanisms

A fired trap produces one possible short-range attempt and launches exactly one ball.

Let q_s be the probability that the short-range nudge succeeds, conditional on an unfired trap occupying the target site. If one side is selected at random, the attempt is directed left or right with probability $q_s/2$ each.

Let $K(d)$ be the probability that the ball lands d sites from its source, where

$$\sum_{d \neq 0} K(d) = 1. \quad (2)$$

Let q_ℓ be the probability that the ball fires an unfired trap after landing on it. Conditional on a suitable trap being present, the probability of a successful long-range attempt at displacement d is $q_\ell K(d)$.

3 Calculate the early cascade condition

Before accounting for target occupancy, the mean number of successful activation attempts made by one fired trap is

$$\kappa = q_s + q_\ell. \quad (3)$$

At the beginning of the cascade, a target contains an unfired trap with probability approximately ρ . Hence the expected number of newly activated traps produced by one fired trap is

$$C_0 = \rho\kappa = \rho(q_s + q_\ell). \quad (4)$$

The cascade tends to shrink for $C_0 < 1$ and can initially grow for $C_0 > 1$. The corresponding onset estimate is

$$\boxed{\rho_{\text{onset}} = \frac{1}{\kappa} = \frac{1}{q_s + q_\ell}}. \quad (5)$$

This is an estimate for the onset of a macroscopic cascade, not the density for complete activation.

4 Introduce density variables

At generation n , define the following densities per lattice site:

Symbol	Meaning
$\rho_{u,n}$	Density of occupied, unfired traps
$\rho_{a,n}$	Density of traps newly activated in generation n
$\rho_{f,n}$	Density of traps that have already fired

Condition on the initially dropped ball activating one trap. A convenient initialization is

$$\rho_{a,0} = \frac{1}{N}, \quad \rho_{u,0} = \rho - \frac{1}{N}, \quad \rho_{f,0} = 0. \quad (6)$$

5 Iterate the mean-field cascade

Assume that activation attempts are independent and uniformly distributed over the sites. The probability that an unfired trap receives at least one attempt from generation n is approximated by $1 - \exp(-\kappa\rho_{a,n})$. Therefore,

$$\rho_{a,n+1} = \rho_{u,n} [1 - \exp(-\kappa\rho_{a,n})], \quad (7)$$

$$\rho_{u,n+1} = \rho_{u,n} - \rho_{a,n+1}, \quad (8)$$

$$\rho_{f,n+1} = \rho_{f,n} + \rho_{a,n}. \quad (9)$$

Iterate until $\rho_{a,n}$ is below a numerical tolerance, for example 10^{-10} . The limiting fired density is denoted $\rho_{f,\infty}$.

6 Calculate the final fired percentage

The initial trap density is ρ . The final fraction and percentage of traps fired are therefore

$$F(\rho) = \frac{\rho_{f,\infty}}{\rho}, \quad \text{percentage fired} = 100F(\rho)\%. \quad (10)$$

7 Solve the direct final-size equation

Under the same homogeneous-mixing and independent-attempt assumptions, the final fired fraction satisfies the self-consistency equation

$$\boxed{F = 1 - \exp(-\rho\kappa F) = 1 - \exp[-\rho(q_s + q_\ell)F]}. \quad (11)$$

It can be solved with a numerical root finder. Fixed-point iteration,

$$F_{m+1} = 1 - \exp(-\rho\kappa F_m), \quad (12)$$

can also be used from a small positive starting value, although it may converge slowly near the threshold.

Origin of the final-size equation

If a fraction F of the initially present traps fires, the fired-trap density is ρF . Each fired trap produces κ attempts on average, so the mean number of attempts received by one site is

$$\lambda = \kappa \rho F. \quad (13)$$

With independent, uniformly distributed attempts, the number received by a given site is approximated by a Poisson random variable. Thus the probability of receiving no attempt is $e^{-\lambda}$, and the probability of receiving at least one is $1 - e^{-\lambda}$. Conditional on a trap being initially present, this latter probability is its probability of firing. Since F is the final fraction fired, self-consistency requires

$$F = 1 - e^{-\lambda} = 1 - e^{-\rho \kappa F}, \quad (14)$$

which is Eq. (11).

8 Scan the density

Evaluate $F(\rho)$ over $0 \leq \rho \leq 1$. Record the smallest density at which the predicted fired fraction reaches 90%, 95%, 99%, and 100%.

For a target $F_* < 1$, Eq. (11) can be rearranged to give

$$\boxed{\rho_* = -\frac{\ln(1 - F_*)}{\kappa F_*}}. \quad (15)$$

Target	Required density	Status condition
90%	$2.5584/\kappa$	attainable only if $\rho_* \leq 1$
95%	$3.1534/\kappa$	attainable only if $\rho_* \leq 1$
99%	$4.6517/\kappa$	attainable only if $\rho_* \leq 1$
100%	no finite value	exponential term remains positive

Since each trap makes at most one nudge attempt and one ball attempt, $\kappa \leq 2$. Consequently, this homogeneous-mixing approximation cannot reach the 90%, 95%, or 99% targets within $0 \leq \rho \leq 1$. This is a limitation of the approximation, not a claim that a spatial chain can never fire completely.

9 Report the model outputs

The model produces:

1. the onset estimate $\rho_{\text{onset}} = 1/\kappa$;
2. the predicted curve $100F(\rho)\%$;
3. the calculated or unattainable status of the 90%, 95%, 99%, and 100% targets.

The assumptions are independent occupancy, homogeneous mixing, independent activation attempts, one ball per fired trap, and no re-firing.